Assignment 1: **Subject: MPCA**

1. Differentiate between Microprocessor and Microcontroller.
2. Differentiate between Von-Neumann and Harvard Architecture with suitable block diagrams.
3. Explain the phases in instruction cycle.
4. Explain the basic computer organization.
5. Construct a 8x4 ROM from two 8x2 ROMs. Also explain it.
6. Design a 16x4 memory subsystem with high order interleaving using 8x2 memory chips for a computer system with an 8-bit address bus.
7. Design a 16x4 memory subsystem with low order interleaving using 8x2 memory chips for a computer system with an 8-bit address bus.
8. Design an interface for an input device which has binary address 1010 1010. Its computer system uses isolated I/0.
9. Design an interface for an input device which has binary address 1010 1010. Its computer system uses memory mapped I/O.
10. Design an interface for a bidirectional Input/output device that has binary address 1000 0001. Its computer system uses Isolated I/O.
11. A CPU with 8-bit data bus and 8-bit address bus uses memory mapped I/O. It has 32 byte of ROM at 00H and 32 byte of ROM at 20H. It also has an I/O device at FFH. Show the load logic and enable logic.
12. A computer system with an 8-bit address and an 8-bit data bus using isolated I/O. It has 16 x 8 ROM starting at address 00H constructed using 8 x8 chips; 64 x 8 of RAM starting at address 80H constructed using 64 x 4 chips. There is an I/O device at 40H. Show the design for the system.
13. 8. A computer system with 8-bit data bus and 8-bit control bus uses isolated I/O. It has 64 bytes of ROM starting at 00H constructed using 64 x 4 chips; 128 bytes of RAM, starting at address 40H constructed using 32 x 8 chips; an input device with READY signal at address 40H; and an output with no READY signal at address 80H. Show the design for this system. Include all enable and load logic.
14. A computer has a CPU with 8-bit address bus and 16-bit data bus. The computer uses Isolated I/O. It has 64 x 16 of ROM at 00 H; constructed using two 32 x 16

ROM chips. It also has 32 x16 of RAM at C0 H. The system has an input device at 17 H and an output device at B5 H. Show the design for the system including all the required logic. [PU 2018 Fall]

15.

A computer system with an 8-bit address bus and an 8-bit data bus uses memory mapped I/O. It has 32 bytes of ROM starting at address 00H constructed using a single 32 × 8 chip; 128 bytes of RAM starting at address 80H constructed using 64 × 4 chips; a bidirectional input/output device with no *READY* signal at address 40H; and an input device with no *READY* signal at address 60H. Show the design for this system. Include all enable and load logic.

16. Design a CPU for below:

- It can access 64 words of memory, each word being 8 bits wide. The CPU does this by outputting a 6-bit address on its output pins A[5..0] and reading in the 8-bit value from memory on its inputs D[7..0].
- The CPU contains a 6-bit address register (*AR*) and program counter (*PC*); an 8-bit accumulator (*AC*) and data register (*DR*); and a 2-bit instruction register (*IR*).
- The CPU must realize the following instruction set.

Instruction	Instruction Code	Operation
COM	00XXXXXX	$AC \leftarrow AC'$
JREL	01AAAAAA	$PC \leftarrow PC + 00AAAAAA$
OR	10AAAAAA	$AC \leftarrow AC \vee M[00AAAAAA]$
SUB1	11AAAAAA	$AC \leftarrow AC - M[00AAAAAA] - 1$

17. Design the hardwired CU for question no 16

18. Design a microsequencer control unit with horizontal microcode for the CPU described in question no. 16

19. Design a microsequencer control unit with vertical microcode for the CPU described in question no. 16

20. Design a microsequencer control unit which directly generate control signals for the CPU described in question no. 16

21. There is a very simple CPU for the given set of Instruction:

Instruction	Instruction Code	Operations
STR	00 AAAAAA	AC→M[AAAAAA]
NAND	01 AAAAAA	AC←(AC^M[AAAAAA])'
JMP	10 AAAAAA	PC← AAAAAA
INC2	11 XXXXXX	AC←AC+2

Let the instruction width be 8 bits and address is 6 bits. Design the CPU's register section, state diagram and ALU

22. Design the hardwired control unit for the CPU described in above question.
23. Design a microsequencer control unit with horizontal microcode for the CPU described in question no. 21
24. Design a microsequencer control unit with vertical microcode for the CPU described in question no. 21
25. Design a microsequencer control unit which directly generate control signals for the CPU described in question no. 21
26. Trace RTL code to multiply 3x5 using shift add algorithm.
27. Trace RTL code to multiply -7 x -9 using Booth's algorithm.
28. Explain memory hierarchy.
29. Explain Associative memory with its block diagram.
30. Explain different cache mapping techniques.

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